Load Balancer for Replicated 5G Core Network Functions

Advanced Computer Networks: Course Project

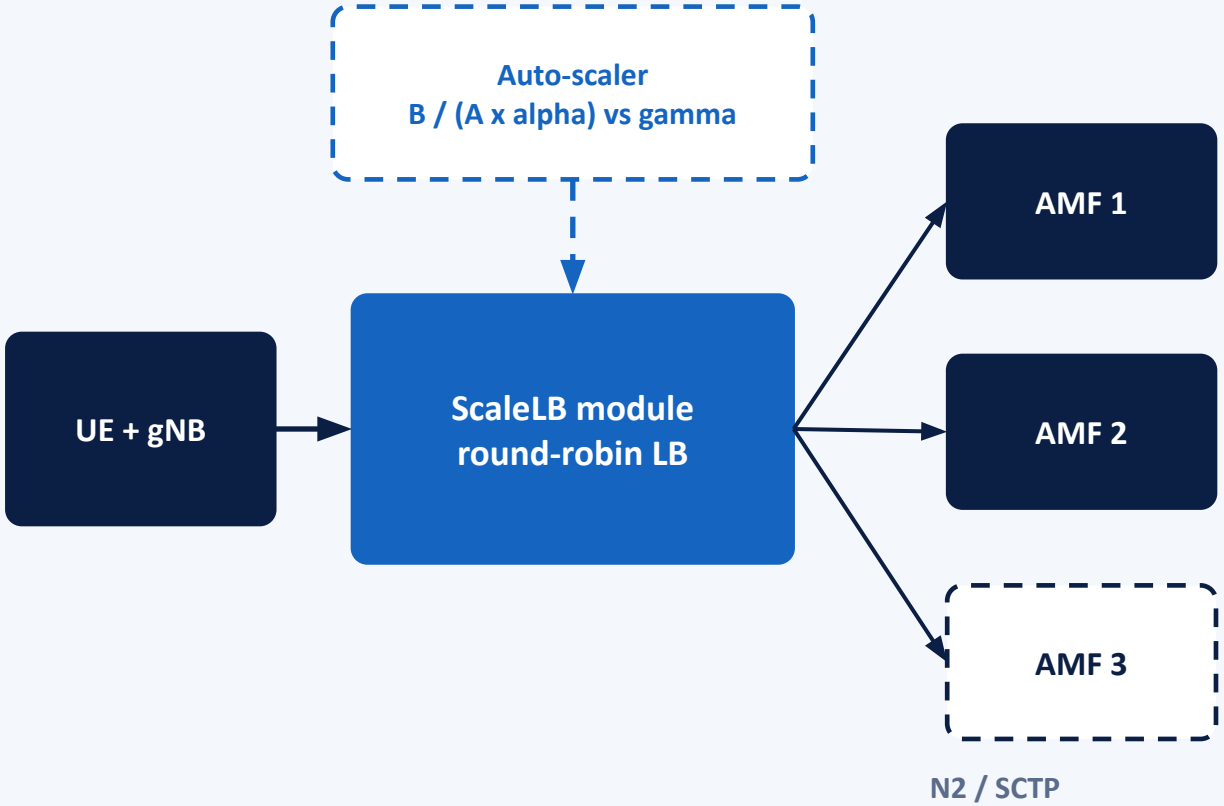
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Base Paper

Dev, Bera and Tiwari, "Control-Plane Load Balancing and Auto-Scaling in 5G and Beyond Networks", ICOIN 2025

What it does

- An auto-scaling load balancer sits between the RAN and the AMFs and hides the core from gNBs
- Round-robin spreads control messages over the AMFs
- Proactive scaling: add an AMF when B / (A x alpha) >= gamma
- Reactive scaling: add an AMF when B / (A x alpha) > 1
- B = active gNBs, A = active AMFs, alpha = gNBs per AMF
- Built on Open5GS, my5G-RANTester and LoxiLB in Kubernetes



Dashed AMF = started by the scaler on demand

What ScaleLB Shows, and What It Leaves Open

Gaps we address

Scaling is threshold-based

It never predicts load; the paper names ML prediction as future work.

Our answer: **LSTM predictive auto-scaler**

No traffic priority

Critical signalling waits behind everything during AMF boot.

Our answer: **Slice-aware priority queuing**

Objectives

1

Slice-aware priority queuing

Protect URLLC signalling from mMTC bursts: URLLC bypasses the queue; mMTC is buffered and rate-limited when AMF CPU exceeds 80%.

2

LSTM-based predictive auto-scaling

Forecast the request rate 45-60 s ahead and boot AMFs early to hide the ~40 s start-up lag.

Feature 1: Slice-Aware Priority Queuing

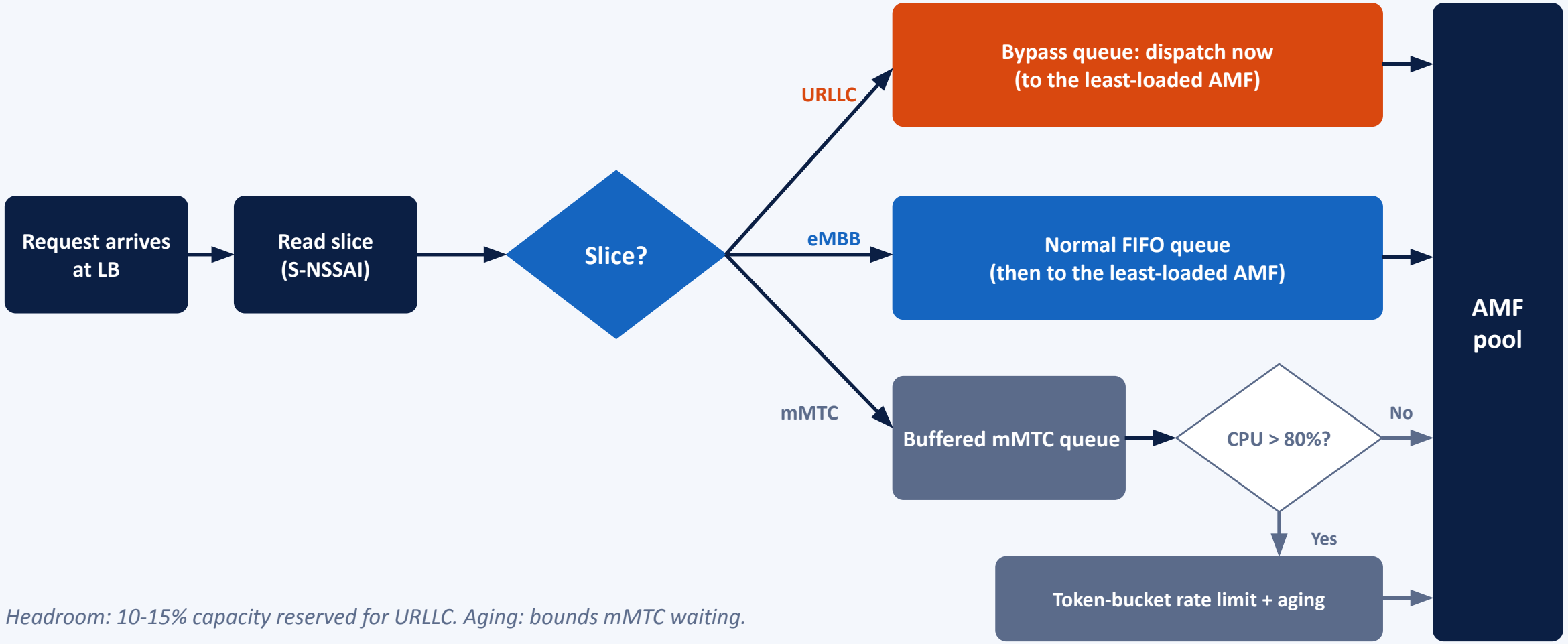
- The load balancer checks which slice each request belongs to
- Some AMF capacity (10-15%) is kept free just for urgent URLLC requests
- When AMFs are more than 80% busy, mMTC requests are released slowly, a few at a time
- If an mMTC request waits too long, it gets pushed forward so it is never ignored

URLLC Bypasses the queue and is dispatched immediately: hashed AMF, else the least-loaded one.

eMBB Normal queue; goes to the next AMF on the hash ring if its hashed AMF is full.

mMTC Buffered queue; rate-limited above 80% CPU; aged so it is eventually served.

Feature 1: Methodology Flow

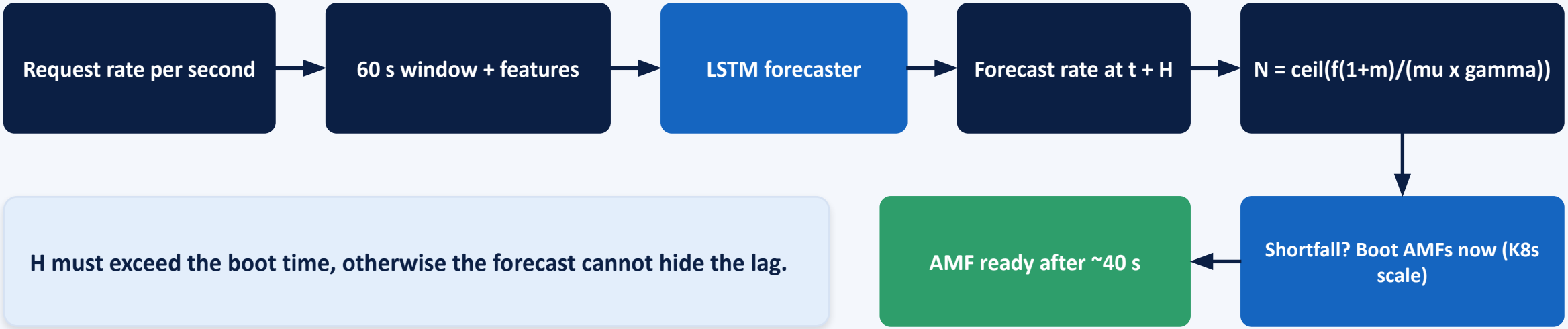


Headroom: 10-15% capacity reserved for URLLC. Aging: bounds mMTC waiting.

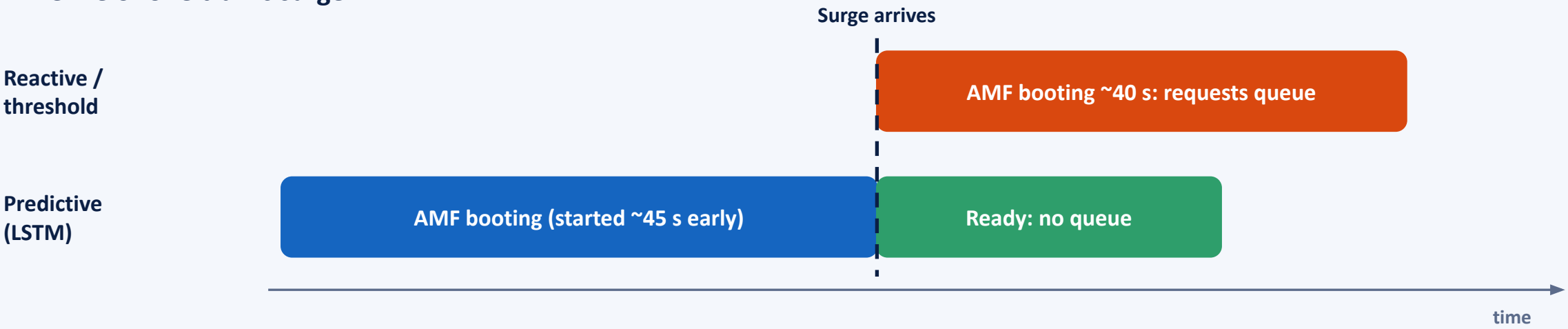
Part 2: LSTM Predictive Auto-Scaling

- A new AMF takes ~40 s to start, so scaling after the load arrives makes requests wait (ScaleLB Fig. 5)
- Our idea: predict the load in advance and start AMFs early
- The LSTM learns from the last 60 s of request rate
- It predicts the load 45-60 s ahead, longer than the 40 s start time
- If more AMFs are needed, they are started now, so they are ready before the load arrives
- Scale down slowly, after AMFs finish their current work
- Compared with: EWMA, ARIMA, Random Forest, reactive scaling

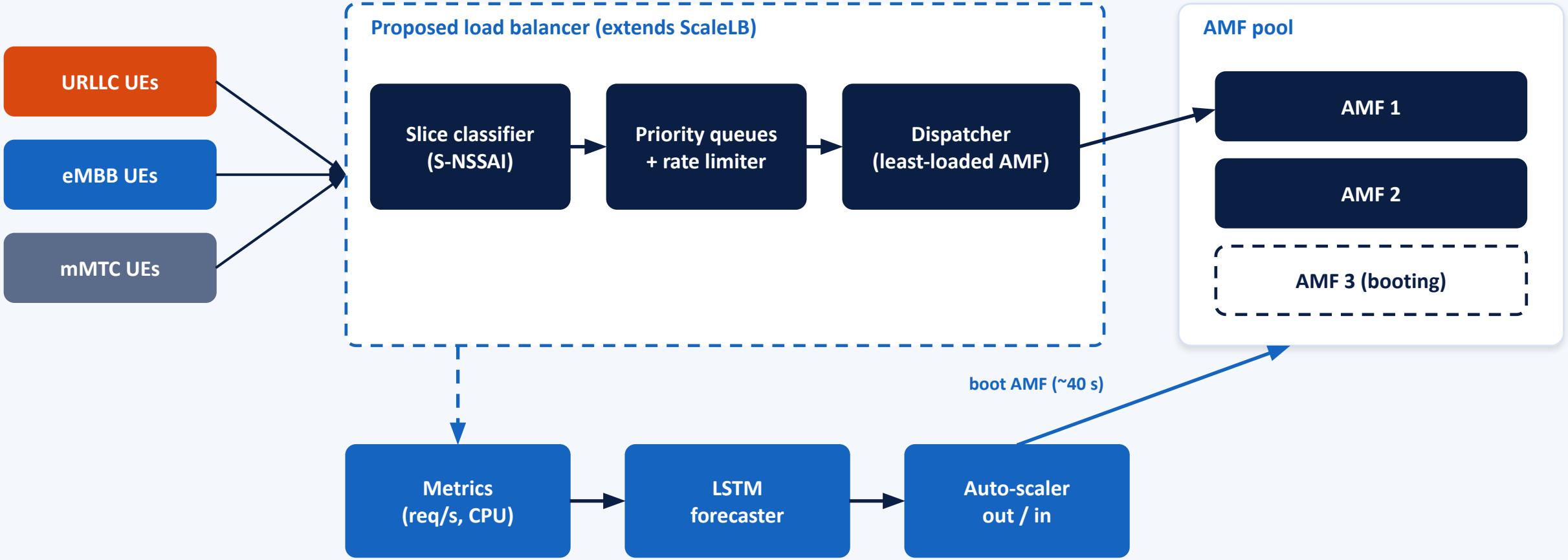
Part 2: Methodology Flow



Timeline of one traffic surge



Proposed System Architecture



References

[1] W. Dev, S. Bera and A. Tiwari, "Control-Plane Load Balancing and Auto-Scaling in 5G and Beyond Networks," ICOIN, 2025, pp. 83-87.

[2] Open5GS: https://open5gs.org/ | my5G-RANTester, UERANSIM, LoxiLB (open-source projects)